Meeting 8 – Part 1

Machine Capability & Hardware Resource Architecture

Project: Digital Harmonium Workstation
Version: Final Discussion Draft
Status: Approved Design Foundation

8.1 Purpose

Before software architecture can be finalized, it is essential to understand the physical capabilities of the target hardware.

A software architecture can only be considered correct if it respects the limitations and strengths of the underlying processor.

Therefore, this meeting establishes the engineering baseline upon which every future software decision will be made.

The objective is not merely to list specifications, but to understand how the ESP32 will behave as a real-time digital musical instrument.

8.2 Design Philosophy

The Digital Harmonium Workstation is designed around a simple philosophy:

> **The hardware must never limit musical expression.**

Instead of designing software around limitations, we first identify every capability of the processor and then exploit those capabilities efficiently.

Every architectural decision made in later meetings assumes the hardware described in this document.

8.3 Selected Processing Platform

The project uses the **ESP32-WROOM development board** as its primary processing platform.

The ESP32 was selected because it provides:

- Dual-core processing
- Hardware floating-point support
- Real-time operating system support
- High-speed I^2S audio interface
- Sufficient Flash memory
- Adequate SRAM
- Integrated SPI peripherals
- Excellent cost-to-performance ratio

Most importantly, the ESP32 allows the project to remain affordable while still supporting professional software architecture.

8.4 Processor Characteristics

Processor Architecture:

- Dual-core Tensilica Xtensa LX6
- 32-bit architecture

Maximum Clock Frequency:

- **240 MHz**

Clock Scaling:

- Dynamic frequency scaling supported
- Lower frequencies may be used for future power-saving modes

The processor provides sufficient computational capability for:

- Multiple simultaneous voices
- Digital synthesis
- Audio mixing
- LCD management
- Recording
- Background storage operations

without requiring dedicated DSP hardware.

8.5 Real-Time Operating System

The workstation will use **FreeRTOS**, which is integrated into the ESP32 software environment.

FreeRTOS provides:

- Task scheduling
- Priorities
- Queues
- Synchronization
- Timers
- Deterministic execution

This allows the workstation to behave as several independent subsystems running simultaneously while maintaining strict real-time behavior.

8.6 Available Memory Resources

The ESP32 provides three major memory resources.

Internal SRAM

Total:

520 KB

Purpose:

- Runtime execution
- Audio buffers
- Voice objects
- FreeRTOS
- Recorder buffers
- Display data

SRAM is reserved for active information only.

Internal Flash

Total:

4 MB

Purpose:

- Firmware

- Operating software
- Active VHL package
- Settings

Flash retains information after power loss.

External SD Card

Purpose:

- Instrument library
- Recordings
- Presets
- User storage
- Future expansion

The SD card functions as permanent archival storage.

8.7 Audio Hardware Architecture

The workstation will employ a dedicated external digital audio chain.

Selected Components:

- **UDA1334A Digital Audio DAC**
- **PAM8403 Stereo Power Amplifier**

Reasons for selection:

- Stereo output
- Better audio quality
- Reduced processor workload
- Clean digital signal path
- Simplified analog circuitry

The ESP32 will transmit digital audio over the I^2S interface directly to the DAC.

The DAC performs digital-to-analog conversion.

The amplifier drives the existing keyboard speakers.

A standard **3.5 mm headphone output** will also be provided.

8.8 Storage Interface

Instrument storage and recording storage will use an SPI-connected SD card.

Reasons:

- Large capacity
- Easy replacement
- Low cost
- Standard filesystem support
- Simple software architecture

The SD card serves as the permanent storage medium rather than active runtime memory.

8.9 Instrument Loading Strategy

A major architectural decision was finalized during this meeting.

Instead of continuously streaming instrument data from the SD card during performance, the selected instrument will first be installed into the ESP32's internal Flash memory.

Workflow:

```
Instrument Library
        │
     SD Card
        │
        ▼
Instrument Loader
        │
        ▼
 Internal Flash
        │
        ▼
 Runtime Playback
```

Advantages:

- Reduced runtime latency
- Eliminates SD card access during normal playing
- Cleaner runtime architecture
- Lower RAM usage
- Simplified audio pipeline

Only one active VHL package resides in Flash at a time, while the SD card retains the complete instrument library.

8.10 Floating-Point Processing Strategy

After evaluating processor capability, the project adopts the following DSP philosophy:

All synthesis calculations will use:

Single-precision floating-point arithmetic (`float`)

Reasons:

- Cleaner implementation
- Direct correspondence with mathematical formulations
- Easier debugging
- Easier maintenance
- Easier future expansion

Floating-point calculations continue until the final output stage.

Immediately before audio transmission, samples are converted into 16-bit PCM for the DMA/I^2S subsystem.

This provides an excellent balance between computational efficiency and software clarity.

8.11 Hardware Interfaces

The workstation will utilize the following interfaces.

Inputs:

- Keyboard matrix
- Control buttons
- Existing sliders
- SD card
- USB programming interface

Outputs:

- Stereo audio
- Internal speakers
- 3.5 mm headphone jack
- 16×2 LCD display

- Keyboard LEDs

These interfaces form the complete hardware interaction layer of the software architecture.

8.12 Engineering Constraints

Although the ESP32 is powerful, the following constraints remain fundamental:

- Fixed SRAM
- Fixed Flash
- Real-time deadlines
- Continuous audio output
- No audible glitches
- Predictable execution

Every future software component must respect these constraints.

8.13 Engineering Strengths

The processor provides several significant advantages.

- High computational performance
- Dual-core execution
- Integrated FreeRTOS
- Dedicated I^2S audio
- Hardware floating-point support
- Sufficient Flash for active instruments
- Expandable SD storage

These strengths make the ESP32 particularly suitable for a low-cost professional musical workstation.

8.14 Meeting Conclusions

This meeting established the hardware foundation of the Digital Harmonium Workstation.

The ESP32 possesses sufficient computational capability to execute the complete synthesis architecture without requiring external DSP processors.

The final hardware philosophy can be summarized as follows:

- ESP32 performs all real-time computation.

- Internal SRAM stores active runtime information.
- Internal Flash stores firmware and the currently installed instrument.
- The SD card serves as the permanent instrument library and recording archive.
- Dedicated external audio hardware provides professional-quality stereo output.
- Floating-point DSP ensures mathematical clarity while maintaining adequate performance.

This document serves as the hardware reference for all remaining software architecture decisions.

End of Meeting 8 – Part 1: Machine Capability & Hardware Resource Architecture